Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Orasanu, Judith M. Ph.D.		
Project Title:	Distributed Team Decision Making in Exploration Missions		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035-1000	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	07/01/2001	**End Date:**	07/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fischer, Ute (Georgia Institute of Technology)		
Grant/Contract No.:	NCC 9-58-NBPF.002.04		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project will address the effects of task-related and psychosocial stressors on team problem solving and decision making in situations similar to those that may be encountered in long-duration space missions. Technologies to monitor affective responses of individual team members will be developed. Findings from our work will establish a basis for introducing countermeasures to support effective team interaction and mission success. Five specific research questions will be addressed: (1) What are the impacts of task-related and psychosocial stressors on problem solving, decision making and interactional behaviors in distributed dynamic problem situations? (2) What task and team strategies are most effective in performing collaborative work under stress? (3) How do gender and cultural factors influence decision-making strategies and team interactions? (4) Are there physiological and biomedical indicators that predict individual stress and deteriorating team interactions? (5) Are these measures robust across gender and national culture? Teams of four participants work together on a computer-based dynamic decision-making task. While working on the task, their physiological responses and facial expressions are monitored for signs of mental and emotional stress.		

	Physiological arousal levels and facial affect will subsequently be correlated with individual and team performance on the decision making task. The task used in our studies involves a simulated search and rescue mission in Antarctica. It presents graphical displays of evolving problem scenarios and supports communication among team members via e-mail and voice messages. The simulated mission will enable us to examine collaborative and competitive behaviors among team members during mission planning and during execution of the tasks under a variety of task constraints.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful long-duration space missions will depend on the ability of team members (both space crews and ground controllers) to respond to unanticipated problems and to collaborate effectively under highly stressful conditions. Problem events will require crews to assess the nature of the problem, including time and resources available and the level of threat presented by the situation. Crews also will need to make plans and decisions to manage the risks associated with the situation. In order to do so, crewmembers must share critical information and coordinate their efforts. However, "the history of space explorations has seen many instances of poor interpersonal relations and faulty decision making" (Committee on Space Biology and Medicine, NRC, 1998), leading NASA to conclude that interpersonal difficulties could well threaten the success of long-duration space missions Team interaction may be further complicated as crewmembers become more diverse in terms of skills, training, gender and cultural backgrounds (Connors, Harrison & Akins, 1985). Findings from this research project will provide a basis for understanding factors that may limit effective interaction and problem solving by team members in space and between space and ground crews. Findings will provide a basis for (a) designing or revising procedures, training practices, and technologies to support effective team performance, and (b) developing technologies for monitoring and predicting breakdown of team interactions so that countermeasures may be introduced before team dynamics have deteriorated to the point of threatening mission success.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 05/21/2010)